

## 8.10 Spectral Mapping, Products in Both Orders, Left Eigenvectors

Once the eigenvalues of  $T$  are known, what can we say about the eigenvalues of  $T^2$ , of  $T^2 - 3T + I$ , or of any polynomial in  $T$ ? And when two matrices are multiplied in the two possible orders, how are the eigenvalues of  $\mathbf{AB}$  and  $\mathbf{BA}$  related? This section answers both questions using the triangular forms of Section 7 and the block matrices of Chapter 7. Along the way it settles two debts: Chapter 6 showed  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  only when  $\mathbf{A}$  is invertible, and Chapter 7 proved that vanishing traces of all powers force a matrix to be nilpotent by a long computation, with no explanation of why. We end with left eigenvectors, the eigenvectors of the transpose, which pair with ordinary eigenvectors in a useful way and prepare the Markov chains of Section 11.

### 8.10.1 Eigenvalues of a polynomial in $T$

One half of the answer needs no theory. If  $T\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $T^k\mathbf{v} = \lambda^k\mathbf{v}$  for every  $k$ , and so  $q(T)\mathbf{v} = q(\lambda)\mathbf{v}$  for every  $q \in F[x]$  (Exercise (B3: Eigenvalues of powers and inverses)). So  $q$  **maps eigenvalues of  $T$  to eigenvalues of  $q(T)$** , over every field and in every dimension.

The other half, that **every** eigenvalue of  $q(T)$  arises this way, can fail. Part (c) of Exercise (B3: Eigenvalues of powers and inverses) already met the rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  on  $\mathbb{R}^2$ : it has no real eigenvalue, while  $\mathbf{R}^2 = -\mathbf{I}$  has the eigenvalue  $-1$ . The eigenvalues of  $\mathbf{R}$  that would map to  $-1$ , namely  $\pm i$ , are missing from  $\mathbb{R}$ . So we expect the full statement to need a characteristic polynomial that splits. With that hypothesis we even get the multiplicities, because a triangular form shows all the eigenvalues at once on its diagonal.

*The eigenvalues of  $q(T)$  are the values of  $q$  at the eigenvalues of  $T$ , counted with multiplicity.*

#### Theorem 8.10.1. Spectral Mapping Theorem

Let  $V$  be a finite-dimensional vector space over  $F$  with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ . Suppose  $p_T$  splits over  $F$ , say

$$p_T(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n),$$

where each eigenvalue appears as often as its algebraic multiplicity. Then for every  $q \in F[x]$ ,

$$p_{q(T)}(x) = (x - q(\lambda_1))(x - q(\lambda_2)) \cdots (x - q(\lambda_n)),$$

and consequently

$$\text{spec}(q(T)) = q(\text{spec}(T)) = \{q(\lambda) : \lambda \in \text{spec}(T)\}.$$

In particular this holds for every operator on a non-zero finite-dimensional complex space, and for every  $\mathbf{A} \in M_n(\mathbb{C})$ .

*Idea.* Choose a basis  $\mathcal{B}$  in which  $T$  is upper triangular (Section 7). Its diagonal is  $\lambda_1, \dots, \lambda_n$ . Products and sums of upper triangular matrices are upper triangular, and their diagonals are

computed position by position, so  $q([T]_{\mathcal{B}}^{\mathcal{B}})$  is upper triangular with diagonal  $q(\lambda_1), \dots, q(\lambda_n)$ . The characteristic polynomial of a triangular matrix is read off its diagonal.

*Proof.* By Theorem 8.7.2 there is a basis  $\mathcal{B}$  of  $V$  such that  $\mathbf{U} = [T]_{\mathcal{B}}^{\mathcal{B}}$  is upper triangular. By Theorem 8.7.4, its diagonal entries are  $\lambda_1, \dots, \lambda_n$  in some order; renumbering the  $\lambda_i$ , we may assume  $u_{ii} = \lambda_i$ .

*Claim.* If  $\mathbf{X}, \mathbf{Y} \in M_n(F)$  are upper triangular, then so are  $\mathbf{X} + \mathbf{Y}$ ,  $c\mathbf{X}$  for  $c \in F$ , and  $\mathbf{XY}$ , with diagonal entries  $x_{ii} + y_{ii}$ ,  $cx_{ii}$  and  $x_{ii}y_{ii}$ .

*Proof.* The statements for  $\mathbf{X} + \mathbf{Y}$  and  $c\mathbf{X}$  are entrywise. For the product,  $(\mathbf{XY})_{ij} = \sum_{k=1}^n x_{ik}y_{kj}$  (Definition 0.9.9). If  $i > j$ , every  $k$  has  $k < i$  or  $k \geq i > j$ , so  $x_{ik} = 0$  or  $y_{kj} = 0$ , and the sum is 0. If  $i = j$ , the only term that can be non-zero has  $k \geq i$  and  $k \leq i$ , so  $(\mathbf{XY})_{ii} = x_{ii}y_{ii}$ . ■

By the claim and induction on  $k$ ,  $\mathbf{U}^k$  is upper triangular with diagonal entries  $\lambda_1^k, \dots, \lambda_n^k$  (for  $k = 0$ ,  $\mathbf{U}^0 = \mathbf{I}$ ). Writing  $q = \sum_k a_k x^k$ , the claim again shows that  $q(\mathbf{U}) = \sum_k a_k \mathbf{U}^k$  is upper triangular with diagonal entries  $\sum_k a_k \lambda_i^k = q(\lambda_i)$ . By Corollary 3.6.3,  $[q(T)]_{\mathcal{B}}^{\mathcal{B}} = q(\mathbf{U})$ , so by Definition 6.10.2 and the examples after Definition 6.10.1,

$$p_{q(T)} = p_{q(\mathbf{U})} = \prod_{i=1}^n (x - q(\lambda_i)).$$

For the spectra: by Theorem 8.2.1,  $\mu \in F$  is an eigenvalue of  $q(T)$  if and only if  $p_{q(T)}(\mu) = \prod_i (\mu - q(\lambda_i)) = 0$ , that is, if and only if  $\mu = q(\lambda_i)$  for some  $i$ , since a product of non-zero field elements is non-zero. The  $\lambda_i$  are exactly the eigenvalues of  $T$ , each listed at least once (Theorem 8.2.1). Hence  $\text{spec}(q(T)) = \{q(\lambda) : \lambda \in \text{spec}(T)\}$ . Over  $\mathbb{C}$ ,  $p_T$  always splits by Corollary 5.5.5. ■

The formula for  $p_{q(T)}$  carries more than the spectrum. The algebraic multiplicity of  $\mu$  as an eigenvalue of  $q(T)$  is the number of indices  $i$  with  $q(\lambda_i) = \mu$  (Lemma 5.4.1), so several eigenvalues of  $T$  can merge into one eigenvalue of  $q(T)$ , and their multiplicities add. Combined with Theorem 8.3.3, applied to  $q(T)$ , whose characteristic polynomial splits by the theorem, we get the most used consequence.

### Corollary 8.10.2. Trace and Determinant of a Polynomial in $T$

Under the hypotheses of Theorem 8.10.1, for every  $q \in F[x]$ ,

$$\text{tr } q(T) = q(\lambda_1) + \dots + q(\lambda_n), \quad \det q(T) = q(\lambda_1) \cdots q(\lambda_n).$$

In particular  $\text{tr } T^k = \lambda_1^k + \dots + \lambda_n^k$  for every  $k \in \mathbb{N}$ .

*Proof.* By Theorem 8.10.1,  $p_{q(T)} = \prod_i (x - q(\lambda_i))$  splits. Group equal values: if  $\mu_1, \dots, \mu_s$  are the distinct values among the  $q(\lambda_i)$ , then  $\mu_j$  occurs exactly  $a_{q(T)}(\mu_j)$  times in the list (Lemma 5.4.1). By Theorem 8.3.3 applied to  $q(T)$ ,  $\text{tr } q(T) = \sum_j a_{q(T)}(\mu_j) \mu_j = \sum_i q(\lambda_i)$  and

$\det q(T) = \prod_j \mu_j^{a_{q(T)}(\mu_j)} = \prod_i q(\lambda_i)$ . Take  $q = x^k$  for the last statement. ■

**Example. A Polynomial in a Non-Diagonalizable Matrix**

Let  $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ -3 & 3 & 2 \end{pmatrix} \in M_3(\mathbb{R})$ , for which Exercise (B1: Diagonalize or show impossible) found  $p_{\mathbf{A}} = (x+1)(x-2)^2$  and showed that  $\mathbf{A}$  is not diagonalizable. Without computing any eigenvectors, find  $p_{\mathbf{A}^2}$ ,  $\text{tr } \mathbf{A}^2$  and  $\det \mathbf{A}^2$ . Then find  $p_{q(\mathbf{A})}$  for  $q = x^2 - x - 2$ , and decide whether  $q(\mathbf{A}) = 0$ .

*Solution. The square.*  $p_{\mathbf{A}}$  splits over  $\mathbb{R}$  with the list  $\lambda_1 = -1, \lambda_2 = \lambda_3 = 2$ . By [Theorem 8.10.1](#) with  $q = x^2$ , the list for  $\mathbf{A}^2$  is 1, 4, 4, so

$$p_{\mathbf{A}^2} = (x-1)(x-4)^2, \quad \text{tr } \mathbf{A}^2 = 1 + 4 + 4 = 9, \quad \det \mathbf{A}^2 = 1 \cdot 4 \cdot 4 = 16$$

by [Corollary 8.10.2](#). *Check.* Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} 0 & 4 & 4 \\ -1 & 5 & 4 \\ -3 & 3 & 4 \end{pmatrix},$$

whose trace is  $0 + 5 + 4 = 9$ ; and  $\det \mathbf{A} = (-1) \cdot 2 \cdot 2 = -4$  by Theorem 8.3.3, so  $\det \mathbf{A}^2 = (\det \mathbf{A})^2 = 16$  by Theorem 6.5.4.

*The polynomial q.* Here  $q = (x-2)(x+1)$ , so  $q(-1) = 0$  and  $q(2) = 0$ . The list for  $q(\mathbf{A})$  is 0, 0, 0, and  $p_{q(\mathbf{A})} = x^3$ : the only eigenvalue of  $q(\mathbf{A})$  is 0. Yet  $q(\mathbf{A}) \neq 0$ . Indeed

$$q(\mathbf{A}) = \mathbf{A}^2 - \mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -3 & 3 & 3 \\ -3 & 3 & 3 \\ 0 & 0 & 0 \end{pmatrix},$$

and one checks  $q(\mathbf{A})^2 = 0$ . The spectral mapping theorem sees only eigenvalues, and a non-zero matrix can have 0 as its only eigenvalue. That  $q(\mathbf{A}) \neq 0$  here is no accident: if  $q(\mathbf{A}) = 0$ , the minimal polynomial of  $\mathbf{A}$  would divide (Theorem 8.5.1)  $(x-2)(x+1)$ , which has distinct roots, and  $\mathbf{A}$  would be diagonalizable by Theorem 8.8.2.

**Warning. Without splitting, spectral mapping fails.** For the real rotation  $\mathbf{R}$  above,  $\text{spec}(\mathbf{R}) = \emptyset$  in  $\mathbb{R}$ , so  $q(\text{spec}(\mathbf{R})) = \emptyset$  for  $q = x^2$ , while  $\text{spec}(\mathbf{R}^2) = \text{spec}(-\mathbf{I}) = \{-1\}$ . Over  $\mathbb{C}$  the theorem is restored:  $\text{spec}(\mathbf{R}) = \{i, -i\}$ , both square to  $-1$ , and  $p_{\mathbf{R}^2} = (x+1)^2$ . Also, the theorem is about eigenvalues and algebraic multiplicities only:  $\mathbf{A} = \text{diag}(1, -1)$  has two one-dimensional eigenspaces, while  $\mathbf{A}^2 = \mathbf{I}$  has the single eigenspace  $\mathbb{R}^2$ .

*Quick check.* Let  $\mathbf{A} \in M_2(\mathbb{C})$  have eigenvalues  $i$  and  $-i$ . Find  $p_{\mathbf{A}^2}$  and  $\det(\mathbf{A}^2 + \mathbf{I})$ .

*Solution.* The list for  $\mathbf{A}$  is  $i, -i$  (two distinct roots of the degree-2 polynomial  $p_{\mathbf{A}}$ ). By Theorem 8.10.1 with  $q = x^2$ , the list for  $\mathbf{A}^2$  is  $-1, -1$ , so  $p_{\mathbf{A}^2} = (x + 1)^2$ . With  $q = x^2 + 1$ , Corollary 8.10.2 gives  $\det(\mathbf{A}^2 + \mathbf{I}) = q(i)q(-i) = 0 \cdot 0 = 0$ .

## 8.10.2 Traces of powers detect nilpotency

Chapter 7 proved, in Exercise (C2: When  $\mathbf{A}$  commutes with  $[\mathbf{A}, \mathbf{B}]$ ), that a matrix  $\mathbf{C}$  over a field of characteristic 0 with  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$  is nilpotent, by a careful argument with annihilating polynomials that gave no reason for the result. Here is the view through eigenvalues, for complex matrices. The traces of the powers are the **power sums**  $\lambda_1^k + \cdots + \lambda_n^k$  of the eigenvalues (Corollary 8.10.2), and a nilpotent matrix is one whose eigenvalues are all 0. So the question becomes: if all power sums of  $n$  complex numbers vanish, must the numbers vanish? A Vandermonde determinant says yes.

### Theorem 8.10.3. Nilpotent Matrices via Eigenvalues and Traces

Let  $n \geq 1$  and  $\mathbf{A} \in M_n(\mathbb{C})$ . The following are equivalent:

- (a)  $\mathbf{A}$  is nilpotent, that is,  $\mathbf{A}^s = 0$  for some  $s \geq 1$ ;
- (b)  $\text{spec}(\mathbf{A}) = \{0\}$ ;
- (c)  $p_{\mathbf{A}} = x^n$ ;
- (d)  $\mathbf{A}^n = 0$ ;
- (e)  $\text{tr } \mathbf{A}^k = 0$  for  $k = 1, 2, \dots, n$ .

The same holds for  $\mathbf{A} \in M_n(F)$  when  $F$  is a subfield of  $\mathbb{C}$ , such as  $\mathbb{Q}$  or  $\mathbb{R}$ , with  $\text{spec}(\mathbf{A})$  in (b) taken over  $\mathbb{C}$ .

*Idea. Plan:* (a)  $\Rightarrow$  (b)  $\Rightarrow$  (c)  $\Rightarrow$  (d)  $\Rightarrow$  (a), and (c)  $\Rightarrow$  (e)  $\Rightarrow$  (b). All but one arrow are short. For (e)  $\Rightarrow$  (b), suppose some eigenvalue is non-zero. Group the non-zero eigenvalues into distinct values  $\mu_1, \dots, \mu_r$  with multiplicities  $m_1, \dots, m_r \geq 1$ . The vanishing traces say  $\sum_j m_j \mu_j^k = 0$  for  $k = 1, \dots, r$ : a square linear system for the vector  $(m_1, \dots, m_r)$ , whose matrix is a Vandermonde matrix times the invertible  $\text{diag}(\mu_j)$ . So the  $m_j$  would be 0, but they are positive integers, and positive integers are non-zero in  $\mathbb{C}$ .

*Proof.*

- (a)  $\Rightarrow$  (b). Let  $\mathbf{A}^s = 0$  with  $s \geq 1$ . If  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  with  $\mathbf{v} \neq \mathbf{0}$ , then  $\mathbf{A}^s\mathbf{v} = \lambda^s\mathbf{v}$  by induction on  $s$ , so  $\lambda^s\mathbf{v} = \mathbf{0}$  and  $\lambda^s = 0$  by Theorem 1.1.9; hence  $\lambda = 0$ , since a product of non-zero scalars is non-zero (Theorem 0.7.1 (f)). So  $\text{spec}(\mathbf{A}) \subseteq \{0\}$ . Conversely  $\mathbf{A}$  is not injective: if it were, its  $s$ -fold composite  $\mathbf{A}^s$  would be injective too (Theorem 0.4.3 (a)), whereas  $\mathbf{A}^s = 0$  is not injective on  $\mathbb{C}^n \neq \{0\}$ . Hence  $0 \in \text{spec}(\mathbf{A})$  by Theorem 8.2.1 ((b)  $\Rightarrow$  (a)), and  $\text{spec}(\mathbf{A}) = \{0\}$ .
- (b)  $\Rightarrow$  (c). By Corollary 5.5.5,  $p_{\mathbf{A}} = (x - z_1) \cdots (x - z_n)$  with  $z_i \in \mathbb{C}$ . Each  $z_i$  is an eigenvalue (Theorem 8.2.1), hence 0, so  $p_{\mathbf{A}} = x^n$ .
- (c)  $\Rightarrow$  (d). By Corollary 8.7.3,  $\mathbf{A} = \mathbf{P}\mathbf{U}\mathbf{P}^{-1}$  with  $\mathbf{P}$  invertible and  $\mathbf{U}$  upper triangular, and by Theorem 8.7.4 the diagonal entries of  $\mathbf{U}$  are the roots of  $p_{\mathbf{A}} = x^n$ , all 0. Then

$\mathbf{U}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{U}\mathbf{e}_j \in \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_{j-1})$  for  $j \geq 2$ , since column  $j$  of  $\mathbf{U}$  has zeros from row  $j$  down. By induction on  $j$ ,  $\mathbf{U}^j \mathbf{e}_j = \mathbf{0}$ , hence  $\mathbf{U}^n \mathbf{e}_j = \mathbf{U}^{n-j} \mathbf{U}^j \mathbf{e}_j = \mathbf{0}$  for every  $j$ , and  $\mathbf{U}^n = \mathbf{0}$ . By Proposition 3.7.6 (c),  $\mathbf{A}^n = \mathbf{P}\mathbf{U}^n\mathbf{P}^{-1} = \mathbf{0}$ .

(d)  $\Rightarrow$  (a). Take  $s = n$ .

(e)  $\Rightarrow$  (e). The list of eigenvalues of  $\mathbf{A}$  with multiplicity is  $0, \dots, 0$ , so by Corollary 8.10.2,  $\text{tr } \mathbf{A}^k = 0^k + \dots + 0^k = 0$  for every  $k \geq 1$ .

(f)  $\Rightarrow$  (b). By Corollary 5.5.5,  $p_{\mathbf{A}}$  splits; let  $\lambda_1, \dots, \lambda_n$  be the list of its roots with multiplicity. Since  $n \geq 1$ ,  $\text{spec}(\mathbf{A})$  is non-empty, so it suffices to show that every  $\lambda_i = 0$ . Suppose not, let  $\mu_1, \dots, \mu_r$  ( $1 \leq r \leq n$ ) be the distinct non-zero values among the  $\lambda_i$ , and let  $m_j \geq 1$  be the number of indices  $i$  with  $\lambda_i = \mu_j$ . For  $1 \leq k \leq r$ , the zero eigenvalues contribute  $0^k = 0$ , so by Corollary 8.10.2 and (e),

$$\sum_{j=1}^r m_j \mu_j^k = \text{tr } \mathbf{A}^k = 0.$$

In matrix form,  $\mathbf{W}\mathbf{m} = \mathbf{0}$ , where  $\mathbf{m} = (m_1, \dots, m_r) \in \mathbb{C}^r$  and  $\mathbf{W} \in M_r(\mathbb{C})$  has  $(k, j)$ -entry  $\mu_j^k = \mu_j^{k-1} \mu_j$ . Hence  $\mathbf{W} = V(\mu_1, \dots, \mu_r)^\top \text{diag}(\mu_1, \dots, \mu_r)$ , where  $V(\mu_1, \dots, \mu_r)$  is the Vandermonde matrix with  $(j, k)$ -entry  $\mu_j^{k-1}$  (Theorem 6.8.1). The  $\mu_j$  are distinct, so  $V$  is invertible (Corollary 6.8.2), and so is  $V^\top$ , with inverse  $(V^{-1})^\top$  (Theorem 0.9.5). The  $\mu_j$  are non-zero, so  $\text{diag}(\mu_1, \dots, \mu_r)$  is invertible. Therefore  $\mathbf{W}$  is invertible, and  $\mathbf{m} = \mathbf{W}^{-1}\mathbf{0} = \mathbf{0}$ . But  $m_1$  is a positive integer, and  $m_1 \neq 0$  in  $\mathbb{C}$ . This contradiction shows that every  $\lambda_i = 0$ , so  $\text{spec}(\mathbf{A}) = \{0\}$ .

For a subfield  $F \subseteq \mathbb{C}$ , regard  $\mathbf{A}$  as a complex matrix. Its powers, their traces, its characteristic polynomial and the condition  $\mathbf{A}^s = \mathbf{0}$  are computed by the same arithmetic in  $F$  and in  $\mathbb{C}$ , so (a), (c), (d) and (e) mean the same thing over  $F$  and over  $\mathbb{C}$ , and the equivalences carry over. ■

In the setting of Exercise (C2: When  $\mathbf{A}$  commutes with  $[\mathbf{A}, \mathbf{B}]$ ), where  $\mathbf{C} = [\mathbf{A}, \mathbf{B}]$  commutes with  $\mathbf{A}$ , part (a) of that exercise gives  $\text{tr } \mathbf{C}^k = 0$  for all  $k \geq 1$ , and for complex matrices the theorem now shows at once that  $\mathbf{C}^n = \mathbf{0}$ . That exercise remains the route for a general field of characteristic 0.

*Warning. The characteristic matters.* Over  $\mathbb{F}_p$ , the identity  $\mathbf{I}_p \in M_p(\mathbb{F}_p)$  has  $\text{tr } \mathbf{I}_p^k = p \cdot 1 = 0$  for every  $k$ , yet  $\mathbf{I}_p^k = \mathbf{I}_p \neq \mathbf{0}$ . The proof above breaks exactly where it used that the positive integer  $m_1$  is non-zero in the field: here the multiplicity  $m_1 = p$  of the eigenvalue 1 is 0 in  $\mathbb{F}_p$ .

### 8.10.3 AB and BA

Chapter 6, in Exercise (B3:  $\mathbf{AB}$  and  $\mathbf{BA}$ ), showed that  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  for square  $\mathbf{A}$  and  $\mathbf{B}$  when  $\mathbf{A}$  is invertible, because then  $\mathbf{BA} = \mathbf{A}^{-1}(\mathbf{AB})\mathbf{A}$  is similar to  $\mathbf{AB}$ . Without invertibility the two products need not be similar, and for rectangular  $\mathbf{A}$  and  $\mathbf{B}$  they do not even have the same size: if  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times m$ , then  $\mathbf{AB}$  is  $m \times m$  and  $\mathbf{BA}$  is  $n \times n$ . Their characteristic polynomials have degrees  $m$  and  $n$ . Multiplying by powers of  $x$  repairs the degrees, and the identity then holds in general.

**Theorem 8.10.4.  $\mathbf{AB}$  and  $\mathbf{BA}$  Have the Same Non-zero Eigenvalues**

Let  $m, n \geq 1$ ,  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ . Then

$$x^n p_{\mathbf{AB}}(x) = x^m p_{\mathbf{BA}}(x).$$

Consequently:

- (a) for every  $\lambda \neq 0$  in  $F$ ,  $a_{\mathbf{AB}}(\lambda) = a_{\mathbf{BA}}(\lambda)$ ; in particular  $\mathbf{AB}$  and  $\mathbf{BA}$  have the same non-zero eigenvalues;
- (b) if  $m = n$ , then  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ ; and if  $m \geq n$ , then  $p_{\mathbf{AB}} = x^{m-n} p_{\mathbf{BA}}$ .

*Idea.* We want one big matrix that “contains”  $\mathbf{AB}$  and another that “contains”  $\mathbf{BA}$ , and that are similar to each other. Put them in  $(m+n) \times (m+n)$  block form:

$$\mathbf{M} = \begin{pmatrix} \mathbf{AB} & 0 \\ \mathbf{B} & 0 \end{pmatrix}, \quad \mathbf{N} = \begin{pmatrix} 0 & 0 \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}.$$

Both are block triangular, so their characteristic polynomials are  $p_{\mathbf{AB}} \cdot x^n$  and  $x^m \cdot p_{\mathbf{BA}}$ . The invertible block matrix  $\mathbf{S} = \begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix}$  turns one into the other:  $\mathbf{MS}$  and  $\mathbf{SN}$  are both  $\begin{pmatrix} \mathbf{AB} & \mathbf{ABA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}$ .

*Proof.* Let  $\mathbf{M}, \mathbf{N}, \mathbf{S} \in M_{m+n}(F)$  be the block matrices displayed in the Idea, partitioned by  $m, n$  in rows and columns. By Theorem 7.1.1,

$$\begin{aligned} \mathbf{MS} &= \begin{pmatrix} \mathbf{AB} \cdot \mathbf{I}_m + 0 & \mathbf{AB} \cdot \mathbf{A} + 0 \\ \mathbf{B} \cdot \mathbf{I}_m + 0 & \mathbf{B} \cdot \mathbf{A} + 0 \end{pmatrix} = \begin{pmatrix} \mathbf{AB} & \mathbf{ABA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}, \\ \mathbf{SN} &= \begin{pmatrix} 0 + \mathbf{AB} & 0 + \mathbf{A} \cdot \mathbf{BA} \\ 0 + \mathbf{B} & 0 + \mathbf{BA} \end{pmatrix} = \begin{pmatrix} \mathbf{AB} & \mathbf{ABA} \\ \mathbf{B} & \mathbf{BA} \end{pmatrix}, \end{aligned}$$

using associativity (Theorem 0.9.4). So  $\mathbf{MS} = \mathbf{SN}$ . The matrix  $\mathbf{S}$  is invertible: by Theorem 7.1.1,  $\begin{pmatrix} \mathbf{I}_m & \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} \begin{pmatrix} \mathbf{I}_m & -\mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} = \begin{pmatrix} \mathbf{I}_m & -\mathbf{A} + \mathbf{A} \\ 0 & \mathbf{I}_n \end{pmatrix} = \mathbf{I}_{m+n}$ , and Theorem 2.4.5 applies. Hence  $\mathbf{S}^{-1}\mathbf{MS} = \mathbf{N}$ , and  $p_{\mathbf{M}} = p_{\mathbf{N}}$  by Theorem 6.10.3.

Now compute both sides. The matrix  $x\mathbf{I}_{m+n} - \mathbf{M} = \begin{pmatrix} x\mathbf{I}_m - \mathbf{AB} & 0 \\ -\mathbf{B} & x\mathbf{I}_n \end{pmatrix}$  is lower block triangular with square diagonal blocks, so by Theorem 6.8.4, which holds over the commutative ring  $F[x]$  (the remarks before Example (A  $3 \times 3$  Characteristic Polynomial)), and by Theorem 6.4.6 for  $x\mathbf{I}_n$ ,

$$p_{\mathbf{M}}(x) = \det(x\mathbf{I}_m - \mathbf{AB}) \det(x\mathbf{I}_n) = p_{\mathbf{AB}}(x) x^n.$$

Likewise  $x\mathbf{I}_{m+n} - \mathbf{N} = \begin{pmatrix} x\mathbf{I}_m & 0 \\ -\mathbf{B} & x\mathbf{I}_n - \mathbf{BA} \end{pmatrix}$  gives  $p_{\mathbf{N}}(x) = x^m p_{\mathbf{BA}}(x)$ . Since  $p_{\mathbf{M}} = p_{\mathbf{N}}$ , this proves  $x^n p_{\mathbf{AB}} = x^m p_{\mathbf{BA}}$ .

- (a) Let  $\lambda \neq 0$ . Then  $\lambda$  is not a root of  $x^n$ , so  $\text{mult}_\lambda(x^n) = 0$ , and similarly for  $x^m$ . By Theorem 5.4.2,

$$a_{\mathbf{AB}}(\lambda) = \text{mult}_\lambda(x^n p_{\mathbf{AB}}) = \text{mult}_\lambda(x^m p_{\mathbf{BA}}) = a_{\mathbf{BA}}(\lambda).$$

A scalar is an eigenvalue exactly when its algebraic multiplicity is positive (Theorem 8.2.1, Definition 5.4.1), so the non-zero eigenvalues agree.

- (b) If  $m \geq n$ , then  $x^n p_{\mathbf{AB}} = x^n (x^{m-n} p_{\mathbf{BA}})$ , and canceling the non-zero polynomial  $x^n$  (Corollary 0.8.4) gives  $p_{\mathbf{AB}} = x^{m-n} p_{\mathbf{BA}}$ . For  $m = n$  this is  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ . ■

This pays off the promise of Exercise (B3:  $\mathbf{AB}$  and  $\mathbf{BA}$ ):  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$  for **all** square  $\mathbf{A}$  and  $\mathbf{B}$ , invertible or not. Comparing coefficients (Theorem 6.10.2) recovers  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$  and  $\det(\mathbf{AB}) = \det(\mathbf{BA})$  once more.

The eigenvectors travel between the two products as well, and by hand. If  $\mathbf{AB}\mathbf{w} = \lambda\mathbf{w}$  with  $\lambda \neq 0$ , then

$$\mathbf{BA}(\mathbf{B}\mathbf{w}) = \mathbf{B}(\mathbf{AB}\mathbf{w}) = \lambda\mathbf{B}\mathbf{w},$$

and  $\mathbf{B}\mathbf{w} \neq \mathbf{0}$ , since  $\mathbf{A}(\mathbf{B}\mathbf{w}) = \lambda\mathbf{w} \neq \mathbf{0}$ . So  $\mathbf{B}\mathbf{w}$  is an eigenvector of  $\mathbf{BA}$  for the same  $\lambda$ : multiplying an eigenvector of  $\mathbf{AB}$  by  $\mathbf{B}$  produces one of  $\mathbf{BA}$ . For  $\lambda = 0$  the recipe can fail, because  $\mathbf{B}\mathbf{w}$  may be  $\mathbf{0}$  — which is the warning below in another guise.

**A rank-one example.** Let  $\mathbf{u}, \mathbf{v} \in F^n$  with  $n \geq 2$ . Take  $\mathbf{A} = \mathbf{u} \in M_{n \times 1}(F)$  and  $\mathbf{B} = \mathbf{v}^\top \in M_{1 \times n}(F)$ . Then  $\mathbf{AB} = \mathbf{uv}^\top$  is  $n \times n$ , and  $\mathbf{BA} = \mathbf{v}^\top \mathbf{u}$  is the  $1 \times 1$  matrix whose entry is the number  $c = \mathbf{v}^\top \mathbf{u}$ , so  $p_{\mathbf{BA}} = x - c$ . Applying the theorem with its  $m$  equal to our  $n$ , and its  $n$  equal to 1, gives  $x p_{\mathbf{uv}^\top} = x^n(x - c)$ , and canceling  $x$ ,

$$p_{\mathbf{uv}^\top}(x) = x^{n-1}(x - \mathbf{v}^\top \mathbf{u}).$$

For instance, the all-ones matrix  $\mathbf{1}\mathbf{1}^\top \in M_n(\mathbb{R})$ , with  $\mathbf{1} = (1, \dots, 1)$ , has  $p = x^{n-1}(x - n)$ : eigenvalue 0 with multiplicity  $n - 1$ , and eigenvalue  $n$ . Evaluating at  $-1$  recovers Theorem 6.8.5:

$$\begin{aligned} \det(\mathbf{I} + \mathbf{uv}^\top) &= (-1)^n \det(-\mathbf{I} - \mathbf{uv}^\top) = (-1)^n p_{\mathbf{uv}^\top}(-1) \\ &= (-1)^n (-1)^{n-1} (-1 - \mathbf{v}^\top \mathbf{u}) = 1 + \mathbf{v}^\top \mathbf{u}. \end{aligned}$$

*Warning.* The zero eigenvalue behaves differently, and  $\mathbf{AB}$  need not be similar to  $\mathbf{BA}$ . Take  $\mathbf{A} = \mathbf{E}_{12}$  and  $\mathbf{B} = \mathbf{E}_{11}$  in  $M_2(F)$ . Then  $\mathbf{AB} = \mathbf{E}_{12}\mathbf{E}_{11} = \mathbf{0}$  and  $\mathbf{BA} = \mathbf{E}_{11}\mathbf{E}_{12} = \mathbf{E}_{12}$ . Both have  $p = x^2$ , as the theorem demands. But  $g_{\mathbf{AB}}(0) = 2$  while  $g_{\mathbf{BA}}(0) = 1$ , and  $\mathbf{AB} = \mathbf{0}$  is similar only to 0, so  $\mathbf{AB} \not\sim \mathbf{BA}$ . For rectangular matrices even the algebraic multiplicities of 0 differ, by  $|m - n|$ .

*Quick check.* Let  $\mathbf{A} \in M_{2 \times 5}(\mathbb{R})$  and  $\mathbf{B} \in M_{5 \times 2}(\mathbb{R})$  with  $p_{\mathbf{AB}} = x^2 - 3x + 2$ . What is  $p_{\mathbf{BA}}$ , and what are  $\text{tr}(\mathbf{BA})$  and  $\det(\mathbf{BA})$ ?

*Solution.* By Theorem 8.10.4 with  $m = 2$ ,  $n = 5$ ,  $x^5 p_{\mathbf{AB}} = x^2 p_{\mathbf{BA}}$ , so  $p_{\mathbf{BA}} = x^3(x^2 - 3x + 2) =$

$x^5 - 3x^4 + 2x^3$ . By Theorem 6.10.2,  $\text{tr}(\mathbf{BA}) = 3$  (equal to  $\text{tr}(\mathbf{AB})$ ), and  $\det(\mathbf{BA}) = (-1)^5 \cdot 0 = 0$ , as it must be, since  $\mathbf{BA}$  has rank at most  $2 < 5$ .

## 8.10.4 Left eigenvectors

An eigenvector multiplies  $\mathbf{A}$  from the right:  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ . Row vectors can multiply from the left, and in applications, especially Markov chains, the row version is often the one with a meaning.

A left eigenvector is a row that  $\mathbf{A}$  only stretches when it acts from the right of the row.

### Definition 8.10.1. Left Eigenvector

Let  $\mathbf{A} \in M_n(F)$  and  $\lambda \in F$ . A left eigenvector of  $\mathbf{A}$  for  $\lambda$  is a non-zero  $\mathbf{y} \in F^n$  with

$$\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top.$$

To distinguish them, the eigenvectors of Definition 8.2.1 are also called **right eigenvectors**.

In words:  $\mathbf{y}^\top$  is a  $1 \times n$  row,  $\mathbf{y}^\top \mathbf{A}$  is again a row, and we ask that it be a multiple of  $\mathbf{y}^\top$ . For example,  $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$  and  $\mathbf{y} = (1, 1)$  give  $\mathbf{y}^\top \mathbf{A} = \begin{pmatrix} 4 & 4 \end{pmatrix} = 4\mathbf{y}^\top$ , so  $(1, 1)$  is a left eigenvector for 4, whereas the right eigenvectors for 4 are the multiples of  $(2, 3)$  (Example (Eigenvalues and Eigenspaces of a  $2 \times 2$  Matrix)). For a diagonal matrix,  $\mathbf{e}_i$  is both a left and a right eigenvector for the  $i$ -th diagonal entry. And  $\mathbf{y} = \mathbf{0}$  always satisfies the equation, which is why it is excluded, exactly as for right eigenvectors.

**Non-example by minimal change.** For the same  $\mathbf{A}$ , change  $\mathbf{y} = (1, 1)$  to  $\mathbf{y} = (1, -1)$ . Then  $\mathbf{y}^\top \mathbf{A} = \begin{pmatrix} 1 - 3 & 2 - 2 \end{pmatrix} = \begin{pmatrix} -2 & 0 \end{pmatrix}$ , which is not a multiple of  $\begin{pmatrix} 1 & -1 \end{pmatrix}$ : the row is non-zero, but the defining equation  $\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top$  has no solution  $\lambda$ . Note that  $(1, -1)$  is a right eigenvector of  $\mathbf{A}$ , for  $-1$ ; the two notions do not agree.

Transposing turns the definition into one we know.

### Proposition 8.10.5. Left Eigenvectors Are Eigenvectors of the Transpose

Let  $\mathbf{A} \in M_n(F)$  and  $\lambda \in F$ .

- (a)  $\mathbf{y}$  is a left eigenvector of  $\mathbf{A}$  for  $\lambda$  if and only if  $\mathbf{y}$  is an eigenvector of  $\mathbf{A}^\top$  for  $\lambda$ .
- (b)  $\mathbf{A}$  has a left eigenvector for  $\lambda$  if and only if  $\lambda \in \text{spec}(\mathbf{A})$ . Moreover  $p_{\mathbf{A}^\top} = p_{\mathbf{A}}$ , and  $a_{\mathbf{A}^\top}(\lambda) = a_{\mathbf{A}}(\lambda)$ ,  $g_{\mathbf{A}^\top}(\lambda) = g_{\mathbf{A}}(\lambda)$ .

*Proof.*

- (a) By Theorem 0.9.5,  $(\mathbf{y}^\top \mathbf{A})^\top = \mathbf{A}^\top \mathbf{y}$  and  $(\lambda \mathbf{y}^\top)^\top = \lambda \mathbf{y}$ . Since transposing twice gives back the original,  $\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top$  if and only if  $\mathbf{A}^\top \mathbf{y} = \lambda \mathbf{y}$ .
- (b) First,  $p_{\mathbf{A}^\top} = p_{\mathbf{A}}$ : by Theorem 0.9.5,  $(x\mathbf{I} - \mathbf{A})^\top = x\mathbf{I} - \mathbf{A}^\top$ , and transposing does not change a determinant (Theorem 6.4.5, which holds over the commutative ring  $F[x]$  by the remark after it), so

$$p_{\mathbf{A}^\top} = \det(x\mathbf{I} - \mathbf{A}^\top) = \det((x\mathbf{I} - \mathbf{A})^\top) = \det(x\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}.$$

Hence  $a_{\mathbf{A}^\top}(\lambda) = a_{\mathbf{A}}(\lambda)$ , and  $\text{spec}(\mathbf{A}^\top) = \text{spec}(\mathbf{A})$  by Theorem 8.2.1. With (a),  $\mathbf{A}$  has a left eigenvector for  $\lambda$  if and only if  $\lambda \in \text{spec}(\mathbf{A}^\top) = \text{spec}(\mathbf{A})$ . Finally,  $\mathbf{A}^\top - \lambda \mathbf{I} = (\mathbf{A} - \lambda \mathbf{I})^\top$ , which has the same rank as  $\mathbf{A} - \lambda \mathbf{I}$  (Theorem 2.5.5), so by Theorem 2.5.7,  $g_{\mathbf{A}^\top}(\lambda) = n - \text{rank}(\mathbf{A} - \lambda \mathbf{I})^\top = n - \text{rank}(\mathbf{A} - \lambda \mathbf{I}) = g_{\mathbf{A}}(\lambda)$ . ■

*Remark.* For an operator  $T$  on  $V$ , left eigenvectors correspond to eigenvectors of the dual map  $T'$  on  $V^*$  (Chapter 4), whose matrix in the dual bases is the transpose (Theorem 4.4.2): an eigenvector of  $T'$  is a functional  $\varphi \neq 0$  with  $\varphi \circ T = \lambda \varphi$ .

Left and right eigenvectors for the **same** eigenvalue are usually unrelated, as (1, 1) and (2, 3) show. For **different** eigenvalues, though, they are always orthogonal in the sense of the plain product  $\mathbf{y}^\top \mathbf{v}$ .

### Theorem 8.10.6. Left and Right Eigenvectors Are Biorthogonal

Let  $\mathbf{A} \in M_n(F)$ , let  $\mathbf{y}$  be a left eigenvector of  $\mathbf{A}$  for  $\mu$ , and let  $\mathbf{v}$  be a right eigenvector of  $\mathbf{A}$  for  $\lambda$ . If  $\lambda \neq \mu$ , then  $\mathbf{y}^\top \mathbf{v} = 0$ .

*Proof.* Compute the  $1 \times 1$  matrix  $\mathbf{y}^\top \mathbf{A} \mathbf{v}$  in two ways, using associativity (Theorem 0.9.4):

$$\mu \mathbf{y}^\top \mathbf{v} = (\mathbf{y}^\top \mathbf{A}) \mathbf{v} = \mathbf{y}^\top \mathbf{A} \mathbf{v} = \mathbf{y}^\top (\mathbf{A} \mathbf{v}) = \lambda \mathbf{y}^\top \mathbf{v}.$$

Hence  $(\lambda - \mu) \mathbf{y}^\top \mathbf{v} = 0$ . Since  $\lambda - \mu \neq 0$ , this forces  $\mathbf{y}^\top \mathbf{v} = 0$  (Theorem 0.7.1). ■

For the example, (1, 1) is a left eigenvector for 4 and (1, -1) a right eigenvector for -1, and indeed  $1 - 1 = 0$ .

For a diagonalizable matrix the left eigenvectors come for free with the diagonalization, and they give a formula for all powers that we use in Section 11.

### Proposition 8.10.7. Expansion in Left and Right Eigenvectors

Let  $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$  with  $\mathbf{P} \in M_n(F)$  invertible and  $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Let  $\mathbf{v}_1, \dots, \mathbf{v}_n$  be the columns of  $\mathbf{P}$ , and let  $\mathbf{y}_1^\top, \dots, \mathbf{y}_n^\top$  be the rows of  $\mathbf{P}^{-1}$ . Then:

- (a)  $\mathbf{v}_i$  is a right eigenvector and  $\mathbf{y}_i$  a left eigenvector of  $\mathbf{A}$  for  $\lambda_i$ ;
- (b)  $\mathbf{y}_i^\top \mathbf{v}_j = 1$  if  $i = j$  and 0 if  $i \neq j$ ;
- (c) for every  $q \in F[x]$ ,

$$q(\mathbf{A}) = \sum_{i=1}^n q(\lambda_i) \mathbf{v}_i \mathbf{y}_i^\top, \quad \text{in particular} \quad \mathbf{A}^k = \sum_{i=1}^n \lambda_i^k \mathbf{v}_i \mathbf{y}_i^\top \quad (k \in \mathbb{N}).$$

*Proof.*

- (a) From  $\mathbf{AP} = \mathbf{PD}$ , column  $i$  reads  $\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i$  (Section 4), and  $\mathbf{v}_i \neq \mathbf{0}$  because  $\mathbf{P}$  is invertible. From  $\mathbf{P}^{-1}\mathbf{A} = \mathbf{D}\mathbf{P}^{-1}$ , row  $i$  reads  $\mathbf{y}_i^\top \mathbf{A} = \lambda_i \mathbf{y}_i^\top$  (Theorem 0.9.2), and  $\mathbf{y}_i \neq \mathbf{0}$  because  $\mathbf{P}^{-1}$  is invertible.
- (b) The  $(i, j)$ -entry of  $\mathbf{P}^{-1}\mathbf{P} = \mathbf{I}$  is row  $i$  of  $\mathbf{P}^{-1}$  times column  $j$  of  $\mathbf{P}$ , that is,  $\mathbf{y}_i^\top \mathbf{v}_j$ .
- (c) By Theorem 8.4.3,  $q(\mathbf{A}) = \mathbf{P}q(\mathbf{D})\mathbf{P}^{-1}$ . The columns of  $\mathbf{P}q(\mathbf{D})$  are  $q(\lambda_i)\mathbf{v}_i$ , and the rows of  $\mathbf{P}^{-1}$  are  $\mathbf{y}_i^\top$ , so Corollary 7.1.3 gives  $q(\mathbf{A}) = \sum_i q(\lambda_i)\mathbf{v}_i\mathbf{y}_i^\top$ . Take  $q = x^k$ .

■

Part (b) is biorthogonality with a normalization:  $\mathbf{y}_i^\top \mathbf{v}_i = 1$ . Part (c) splits  $\mathbf{A}$  into rank-one pieces  $\mathbf{v}_i\mathbf{y}_i^\top$ , one per eigenvalue, and each piece is simply multiplied by  $\lambda_i^k$  when  $\mathbf{A}$  is raised to the power  $k$ . In the running example,  $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$  and  $\mathbf{D} = \text{diag}(4, -1)$  give  $\mathbf{P}^{-1} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ 3 & -2 \end{pmatrix}$ , so  $\mathbf{y}_1 = \frac{1}{5}(1, 1)$  and  $\mathbf{y}_2 = \frac{1}{5}(3, -2)$ , and

$$\mathbf{A}^k = \frac{4^k}{5} \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} + \frac{(-1)^k}{5} \begin{pmatrix} 3 & -2 \\ -3 & 2 \end{pmatrix}.$$

For  $k = 1$  this is  $\frac{1}{5} \begin{pmatrix} 8-3 & 8+2 \\ 12+3 & 12-2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ , as it should be.

## 8.10.5 Exercises

### 8.10.5.1 A. Check your understanding

#### Exercise. A1

- (a) State the Spectral Mapping Theorem, including its hypothesis on  $p_T$ .
- (b) True or false: for every  $\mathbf{A} \in M_n(\mathbb{R})$ , every real eigenvalue of  $\mathbf{A}^2$  is the square of a real eigenvalue of  $\mathbf{A}$ . Justify your answer.
- (c) True or false: if  $\mathbf{A} \in M_{2 \times 3}(F)$  and  $\mathbf{B} \in M_{3 \times 2}(F)$ , then  $p_{\mathbf{AB}} = p_{\mathbf{BA}}$ . Justify your answer.
- (d) True or false: for all  $\mathbf{A}, \mathbf{B} \in M_n(F)$ ,  $\mathbf{AB}$  is similar to  $\mathbf{BA}$ . Justify your answer.
- (e) Define a left eigenvector of  $\mathbf{A} \in M_n(F)$ , and say how left eigenvectors of  $\mathbf{A}$  are related to  $\mathbf{A}^\top$ .
- (f) True or false: a matrix  $\mathbf{A} \in M_3(\mathbb{C})$  with  $\text{tr } \mathbf{A} = \text{tr } \mathbf{A}^2 = \text{tr } \mathbf{A}^3 = 0$  satisfies  $\mathbf{A}^3 = 0$ . Justify your answer.

*Solution.*

- (a) If  $V$  is finite-dimensional with  $\dim V = n \geq 1$ ,  $T \in \mathcal{L}(V)$ , and  $p_T = \prod_{i=1}^n (x - \lambda_i)$  splits over  $F$ , then for every  $q \in F[x]$ ,  $p_{q(T)} = \prod_i (x - q(\lambda_i))$  and  $\text{spec}(q(T)) = q(\text{spec}(T))$  (Theorem 8.10.1).
- (b) False. The rotation  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has no real eigenvalue, while  $\mathbf{R}^2 = -\mathbf{I}$  has the real eigenvalue  $-1$ .

- (c) False. They have different degrees, 2 and 3. The correct identity is  $x^3 p_{\mathbf{AB}} = x^2 p_{\mathbf{BA}}$ , that is,  $p_{\mathbf{BA}} = x p_{\mathbf{AB}}$  (Theorem 8.10.4).
- (d) False.  $\mathbf{A} = \mathbf{E}_{12}$ ,  $\mathbf{B} = \mathbf{E}_{11}$  give  $\mathbf{AB} = 0$  and  $\mathbf{BA} = \mathbf{E}_{12} \neq 0$ , and the only matrix similar to 0 is 0.
- (e) A non-zero  $\mathbf{y} \in F^n$  with  $\mathbf{y}^\top \mathbf{A} = \lambda \mathbf{y}^\top$  for some  $\lambda \in F$ . Equivalently,  $\mathbf{y}$  is an eigenvector of  $\mathbf{A}^\top$  for  $\lambda$  (Proposition 8.10.5).
- (f) True. By Theorem 8.10.3 with  $n = 3$ , (e)  $\Rightarrow$  (d).

### 8.10.5.2 B. Practice

#### Exercise. B1: Eigenvalues of a polynomial in $\mathbf{A}$

Let  $\mathbf{A} \in M_3(\mathbb{C})$  have  $p_{\mathbf{A}} = (x - 1)^2(x + 2)$ .

- (a) Find  $p_{\mathbf{A}^2 + \mathbf{A} + \mathbf{I}}$ ,  $\text{tr}(\mathbf{A}^2 + \mathbf{A} + \mathbf{I})$  and  $\det(\mathbf{A}^2 + \mathbf{A} + \mathbf{I})$ . Is  $\mathbf{A}^2 + \mathbf{A} + \mathbf{I}$  invertible?
- (b) Find  $p_{\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I}}$ . Hence show that  $\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I}$  is nilpotent.
- (c) Find  $\text{tr} \mathbf{A}^5$  and  $\det(\mathbf{A}^{-1})$ .

*Solution.* The list of eigenvalues of  $\mathbf{A}$  with multiplicity is 1, 1,  $-2$ , since  $p_{\mathbf{A}}$  splits.

- (a) With  $q = x^2 + x + 1$ ,  $q(1) = 3$  and  $q(-2) = 4 - 2 + 1 = 3$ . By Theorem 8.10.1,  $p_q(\mathbf{A}) = (x - 3)^3$ , and by Corollary 8.10.2,  $\text{tr} q(\mathbf{A}) = 9$  and  $\det q(\mathbf{A}) = 27 \neq 0$ . So  $\mathbf{A}^2 + \mathbf{A} + \mathbf{I}$  is invertible (Theorem 6.5.3).
- (b) With  $r = x^2 + x - 2 = (x - 1)(x + 2)$ ,  $r(1) = r(-2) = 0$ , so  $p_r(\mathbf{A}) = x^3$  by Theorem 8.10.1. By Theorem 8.10.3 ((c)  $\Rightarrow$  (d)),  $r(\mathbf{A})^3 = 0$ , so  $\mathbf{A}^2 + \mathbf{A} - 2\mathbf{I}$  is nilpotent.
- (c) By Corollary 8.10.2,  $\text{tr} \mathbf{A}^5 = 1 + 1 + (-2)^5 = -30$ . Also  $\det \mathbf{A} = 1 \cdot 1 \cdot (-2) = -2 \neq 0$  (Theorem 8.3.3), so  $\mathbf{A}$  is invertible and  $\det(\mathbf{A}^{-1}) = (\det \mathbf{A})^{-1} = -\frac{1}{2}$  (Corollary 6.5.6).

#### Exercise. B2: $\mathbf{AB}$ and $\mathbf{BA}$ for a $2 \times 3$ example

Let  $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix}$ .

- (a) Compute  $\mathbf{AB}$ ,  $\mathbf{BA}$ ,  $p_{\mathbf{AB}}$  and  $p_{\mathbf{BA}}$ , and verify that  $x^3 p_{\mathbf{AB}} = x^2 p_{\mathbf{BA}}$ .
- (b) Hence list the eigenvalues of  $\mathbf{BA}$  without further computation, and find an eigenvector of  $\mathbf{BA}$  for each non-zero eigenvalue by applying  $\mathbf{B}$  to eigenvectors of  $\mathbf{AB}$ .

*Solution.*

- (a) By row-times-column,

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 2 & 1 \end{pmatrix}.$$

$\mathbf{AB}$  is upper triangular, so  $p_{\mathbf{AB}} = (x-2)(x+2) = x^2 - 4$ . Expanding  $\det(x\mathbf{I} - \mathbf{BA})$  along the second row, whose only non-zero entry is  $x+2$  in position  $(2, 2)$ ,

$$\begin{aligned} p_{\mathbf{BA}} &= (x+2) \det \begin{pmatrix} x-1 & -1 \\ -1 & x-1 \end{pmatrix} \\ &= (x+2)((x-1)^2 - 1) = (x+2)x(x-2) = x^3 - 4x. \end{aligned}$$

Then  $x^3 p_{\mathbf{AB}} = x^5 - 4x^3 = x^2(x^3 - 4x) = x^2 p_{\mathbf{BA}}$ .

- (b) The eigenvalues of  $\mathbf{BA}$  are the roots of  $x(x-2)(x+2)$ :  $0, 2, -2$ , matching [Theorem 8.10.4](#) (the non-zero ones are those of  $\mathbf{AB}$ ). If  $\mathbf{AB}\mathbf{w} = \lambda\mathbf{w}$ , then  $\mathbf{BA}(\mathbf{B}\mathbf{w}) = \mathbf{B}(\mathbf{AB}\mathbf{w}) = \lambda\mathbf{B}\mathbf{w}$ , so  $\mathbf{B}\mathbf{w}$  is an eigenvector of  $\mathbf{BA}$  as long as  $\mathbf{B}\mathbf{w} \neq \mathbf{0}$ . For  $\lambda = 2$ :  $\mathbf{w} = (1, 0)$ ,  $\mathbf{B}\mathbf{w} = (1, 0, 1)$ , and  $\mathbf{BA}(1, 0, 1) = (2, 0, 2)$ . For  $\lambda = -2$ :  $\mathbf{AB} + 2\mathbf{I} = \begin{pmatrix} 4 & 1 \\ 0 & 0 \end{pmatrix}$  gives  $\mathbf{w} = (1, -4)$ ,  $\mathbf{B}\mathbf{w} = (1, 4, -3)$ , and  $\mathbf{BA}(1, 4, -3) = (1-3, -8, 1+8-3) = (-2, -8, 6) = -2(1, 4, -3)$ .

### Exercise. B3: Left eigenvectors of a $2 \times 2$ matrix

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix} \in M_2(\mathbb{R})$ , whose right eigenvectors  $(1, 2)$  for 2 and  $(1, 3)$  for 3 were found in Exercise (B1: Diagonalize or show impossible).

- Find a left eigenvector of  $\mathbf{A}$  for each eigenvalue.
- Verify [Theorem 8.10.6](#) for these vectors, and rescale the left eigenvectors so that  $\mathbf{y}_i^\top \mathbf{v}_i = 1$ .
- Hence write  $\mathbf{A}^k$  as a combination of two rank-one matrices, and check the formula for  $k = 0$  and  $k = 1$ .

*Solution.*

- By [Proposition 8.10.5](#), we need eigenvectors of  $\mathbf{A}^\top = \begin{pmatrix} 0 & -6 \\ 1 & 5 \end{pmatrix}$ . For 2:  $\mathbf{A}^\top - 2\mathbf{I} = \begin{pmatrix} -2 & -6 \\ 1 & 3 \end{pmatrix}$  gives  $y_1 = -3y_2$ , so  $(3, -1)$ . For 3:  $\mathbf{A}^\top - 3\mathbf{I} = \begin{pmatrix} -3 & -6 \\ 1 & 2 \end{pmatrix}$  gives  $y_1 = -2y_2$ , so  $(2, -1)$ . Check:  $\begin{pmatrix} 3 & -1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 6 & -2 \end{pmatrix}$  and  $\begin{pmatrix} 2 & -1 \end{pmatrix} \mathbf{A} = \begin{pmatrix} 6 & -3 \end{pmatrix}$ .
- $(3, -1) \cdot (1, 3) = 0$  and  $(2, -1) \cdot (1, 2) = 0$ , as the theorem predicts for different eigenvalues. Also  $(3, -1) \cdot (1, 2) = 1$  and  $(2, -1) \cdot (1, 3) = -1$ , so take  $\mathbf{y}_1 = (3, -1)$  and  $\mathbf{y}_2 = (-2, 1)$ .
- With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$ , the rows of  $\mathbf{P}^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$  are exactly  $\mathbf{y}_1^\top, \mathbf{y}_2^\top$ , and by [Proposition 8.10.7](#),

$$\mathbf{A}^k = 2^k \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix} + 3^k \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} -2 & 1 \end{pmatrix} = 2^k \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix} + 3^k \begin{pmatrix} -2 & 1 \\ -6 & 3 \end{pmatrix}.$$

$$\text{For } k = 0: \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \text{ For } k = 1: \begin{pmatrix} 6-6 & -2+3 \\ 12-18 & -4+9 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix} = \mathbf{A}.$$

### 8.10.5.3 C. Going deeper

#### Exercise. C1: Traces of powers of $\mathbf{AB}$ and $\mathbf{BA}$

Let  $\mathbf{A} \in M_{m \times n}(F)$  and  $\mathbf{B} \in M_{n \times m}(F)$ .

- Prove that  $\mathbf{B}(\mathbf{AB})^{k-1}\mathbf{A} = (\mathbf{BA})^k$  for every  $k \geq 1$ , and deduce that  $\text{tr}((\mathbf{AB})^k) = \text{tr}((\mathbf{BA})^k)$ .
- Now let  $F = \mathbb{C}$ . Give a second proof of the trace identity from [Theorem 8.10.4](#) and [Corollary 8.10.2](#).

*Solution.*

- By induction on  $k$ . For  $k = 1$  both sides are  $\mathbf{BA}$ . If  $\mathbf{B}(\mathbf{AB})^{k-1}\mathbf{A} = (\mathbf{BA})^k$ , then by associativity

$$\begin{aligned} \mathbf{B}(\mathbf{AB})^k\mathbf{A} &= \mathbf{B}(\mathbf{AB})^{k-1}(\mathbf{AB})\mathbf{A} = (\mathbf{B}(\mathbf{AB})^{k-1}\mathbf{A})(\mathbf{BA}) \\ &= (\mathbf{BA})^k(\mathbf{BA}) = (\mathbf{BA})^{k+1}. \end{aligned}$$

Now  $(\mathbf{AB})^k = \mathbf{A}(\mathbf{B}(\mathbf{AB})^{k-1})$ , with  $\mathbf{A}$  of size  $m \times n$  and  $\mathbf{B}(\mathbf{AB})^{k-1}$  of size  $n \times m$ . By [Theorem 0.9.6 \(3\)](#),

$$\text{tr}((\mathbf{AB})^k) = \text{tr}(\mathbf{A} \cdot \mathbf{B}(\mathbf{AB})^{k-1}) = \text{tr}(\mathbf{B}(\mathbf{AB})^{k-1} \cdot \mathbf{A}) = \text{tr}((\mathbf{BA})^k).$$

- Over  $\mathbb{C}$ ,  $p_{\mathbf{AB}}$  and  $p_{\mathbf{BA}}$  split. Suppose  $m \geq n$  (otherwise swap the roles of  $\mathbf{A}$  and  $\mathbf{B}$ ). By [Theorem 8.10.4](#),  $p_{\mathbf{AB}} = x^{m-n}p_{\mathbf{BA}}$ , so if  $\lambda_1, \dots, \lambda_n$  is the list of eigenvalues of  $\mathbf{BA}$  with multiplicity, then the list for  $\mathbf{AB}$  is  $\lambda_1, \dots, \lambda_n$  together with  $m - n$  zeros. For  $k \geq 1$  the zeros contribute  $0^k = 0$ , so by [Corollary 8.10.2](#),  $\text{tr}((\mathbf{AB})^k) = \sum_i \lambda_i^k = \text{tr}((\mathbf{BA})^k)$ .

#### Exercise. C2: Adding a commuting nilpotent matrix

Let  $\mathbf{A}, \mathbf{N} \in M_n(\mathbb{C})$ ,  $n \geq 1$ , with  $\mathbf{AN} = \mathbf{NA}$  and  $\mathbf{N}$  nilpotent.

- Prove that  $p_{\mathbf{A}+\mathbf{N}} = p_{\mathbf{A}}$ ; in particular  $\text{spec}(\mathbf{A} + \mathbf{N}) = \text{spec}(\mathbf{A})$  and  $\det(\mathbf{A} + \mathbf{N}) = \det \mathbf{A}$ .
- Deduce that if  $\mathbf{A}$  is invertible, so is  $\mathbf{A} + \mathbf{N}$ .
- Show that (a) fails without  $\mathbf{AN} = \mathbf{NA}$ .

*Hint: triangularize  $\mathbf{A}$  and  $\mathbf{N}$  together.*

*Solution.*

- By [Theorem 8.9.4](#) there is an invertible  $\mathbf{P}$  with  $\mathbf{U} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  and  $\mathbf{U}' = \mathbf{P}^{-1}\mathbf{N}\mathbf{P}$  upper triangular. Since  $\mathbf{N}$  is nilpotent,  $\text{spec}(\mathbf{N}) = \{0\}$  ([Exercise \(C2: Nilpotent operators\)](#)), so every diagonal entry of  $\mathbf{U}'$  is 0 by [Theorem 8.7.4](#). Then  $\mathbf{P}^{-1}(\mathbf{A} + \mathbf{N})\mathbf{P} = \mathbf{U} + \mathbf{U}'$  is upper triangular with the same diagonal as  $\mathbf{U}$ . By [Theorem 6.10.3](#) and the examples

after Definition 6.10.1,

$$p_{\mathbf{A}+\mathbf{N}} = p_{\mathbf{U}+\mathbf{U}'} = \prod_i (x - u_{ii}) = p_{\mathbf{U}} = p_{\mathbf{A}}.$$

Equal characteristic polynomials have the same roots (Theorem 8.2.1) and the same constant term  $(-1)^n \det$  (Theorem 6.10.2).

(b) By (a),  $\det(\mathbf{A} + \mathbf{N}) = \det \mathbf{A} \neq 0$ , so  $\mathbf{A} + \mathbf{N}$  is invertible (Theorem 6.5.3).

(c) Let  $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Both are nilpotent, so  $p_{\mathbf{A}} = x^2$ , but  $\mathbf{A} + \mathbf{N} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has  $p = x^2 - 1$  and eigenvalues  $\pm 1$ . Here  $\mathbf{AN} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{NA}$ .

**Exercise. C3: How many traces, and which field?**

- (a) Let  $\omega = e^{2\pi i/3} \in \mathbb{C}$  and  $\mathbf{C} = \text{diag}(1, \omega, \omega^2)$ . Show that  $\text{tr } \mathbf{C} = \text{tr } \mathbf{C}^2 = 0$  but  $\mathbf{C}$  is not nilpotent. Explain why this does not contradict Theorem 8.10.3.
- (b) Let  $n \geq 2$ . Give a matrix  $\mathbf{C} \in M_n(\mathbb{C})$  with  $\text{tr } \mathbf{C}^k = 0$  for  $k = 1, \dots, n-1$  that is not nilpotent.
- (c) Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  and suppose  $\mathbf{C} = \mathbf{AB} - \mathbf{BA}$  commutes with  $\mathbf{A}$ . Prove that  $\mathbf{C}^n = 0$ .
- (d) Over  $\mathbb{F}_2$ , find a non-nilpotent  $\mathbf{C} \in M_2(\mathbb{F}_2)$  with  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$ .

*Solution.*

- (a)  $\omega^3 = 1$  and  $\omega \neq 1$ , so  $0 = \omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1)$  gives  $1 + \omega + \omega^2 = 0$ . Hence  $\text{tr } \mathbf{C} = 1 + \omega + \omega^2 = 0$ , and  $\text{tr } \mathbf{C}^2 = 1 + \omega^2 + \omega^4 = 1 + \omega^2 + \omega = 0$ . But  $\mathbf{C}^k = \text{diag}(1, \omega^k, \omega^{2k}) \neq 0$  for every  $k$ . No contradiction: the theorem, for  $n = 3$ , requires  $\text{tr } \mathbf{C}^k = 0$  for  $k = 1, 2, 3$ , and  $\text{tr } \mathbf{C}^3 = 1 + 1 + 1 = 3 \neq 0$ .
- (b) Let  $\zeta = e^{2\pi i/n}$  and  $\mathbf{C} = \text{diag}(1, \zeta, \dots, \zeta^{n-1})$ . For  $1 \leq k \leq n-1$ ,  $\zeta^k \neq 1$  and  $(\zeta^k)^n = 1$ , so  $\text{tr } \mathbf{C}^k = \sum_{j=0}^{n-1} (\zeta^k)^j = \frac{(\zeta^k)^n - 1}{\zeta^k - 1} = 0$  by the geometric sum. But  $\mathbf{C}^k$  has the entry 1 in position  $(1, 1)$  for all  $k$ , so  $\mathbf{C}$  is not nilpotent.
- (c) By Exercise (C2: When  $\mathbf{A}$  commutes with  $[\mathbf{A}, \mathbf{B}]$ ) (a),  $\text{tr } \mathbf{C}^k = 0$  for every  $k \geq 1$ , in particular for  $k = 1, \dots, n$ . By Theorem 8.10.3 ((e)  $\Rightarrow$  (d)),  $\mathbf{C}^n = 0$ .
- (d)  $\mathbf{C} = \mathbf{I}_2$ :  $\text{tr } \mathbf{C}^k = 1 + 1 = 0$  in  $\mathbb{F}_2$  for every  $k$ , and  $\mathbf{C}^k = \mathbf{I}_2 \neq 0$ .